Notes: Hoberg, Phillips, and Prabhala (JF, 2014)

Product Market Threats, Payouts, and Financial Flexibility

Contents

1 Big picture

1.1 Question

Hoberg, Phillips, and Prabhala ask whether product market threats shape corporate financial policy. Their central question is not simply whether firms in risky industries pay less. It is whether forward-looking threats embedded in the changing product descriptions of rivals help explain dividends, repurchases, and cash holdings.

The paper focuses on three policies: whether firms pay dividends, whether they repurchase shares, and how much cash they hold. The hypothesis is that firms facing more fluid and threatening product markets preserve financial flexibility: they are less likely to distribute cash and more likely to keep cash on the balance sheet.

1.2 Setting

The sample consists of annual U.S. firm observations from 1997 to 2008. The key new data source is the business description in each firm's 10-K. The authors use computational linguistics to convert product descriptions into vectors of product words, and then use these vectors to measure how much rival firms are changing in product spaces related to the focal firm.

This setting is useful because 10-K product descriptions are updated each year and are legally required to be accurate and material. The text therefore gives a more current and continuous view of product market dynamics than SIC or NAICS codes.

1.3 Core finding

The paper's main result is that product market fluidity is strongly related to conservative financial policy. Firms facing high fluidity are less likely to pay dividends, less likely to repurchase shares, and hold more cash. The cash result is strongest for firms with less access to external capital, such as young firms, loss-making firms, and noninvestment-grade firms.

The results remain after controlling for standard determinants such as size, age, market-to-book, profitability, asset growth, R&D, retained earnings, cash-flow risk, industry concentration, and industry fixed effects.

1.4 Economic takeaway

The paper links product market competition to financial flexibility. Product market threats make future cash flows and investment needs less stable. A firm that pays out too much cash may be less able to respond to entry, innovation, or product-market displacement by rivals. Thus, product market threats can rationally lead firms to retain cash and avoid payout commitments.

2 Incremental contribution of the paper

2.1 Relative to the payout literature

A large payout literature emphasizes firm maturity, profitability, investment opportunities, risk, and agency problems. This paper adds a distinct product-market dimension of maturity. A firm can be old and profitable, but still face a newly fluid product market if rivals are rapidly changing products around it. Conversely, a young firm in a stable product niche may face less threat. The paper therefore separates **firm maturity** from **product market maturity**.

2.2 Relative to cash-holdings papers

Earlier cash-holdings work emphasizes transaction motives, taxes, agency, and precautionary motives. Hoberg, Phillips, and Prabhala add a direct measure of the threats that create precautionary cash demand. Rather than relying only on historical cash-flow volatility, they use forward-looking product text to measure rivals' product movements. This helps explain why firms may hold cash even when realized historical cash-flow risk does not look especially high.

2.3 Relative to static competition measures

The paper moves beyond static measures such as industry concentration. A concentrated market may still be dynamically threatening if rivals are changing products rapidly. A fragmented market may be less threatening if firms' product descriptions are stable. Product market fluidity measures the *change* in rivals' product space, not just the current number or size distribution of rivals.

2.4 Relative to text-based finance

The paper demonstrates that business-description text can be used not only to classify industries, but also to measure product-market dynamics. It extends the Hoberg and Phillips text-based product market approach from static similarity to dynamic threat.

3 Hypotheses and mechanisms

3.1 Hypothesis 1: Product market fluidity and payout policy

The first hypothesis is that competitive threats reflected in product market fluidity are negatively related to payouts. Firms facing more fluid product markets should be less likely to pay dividends or repurchase shares.

The dividend prediction is especially strong because dividends are sticky. Once a firm establishes a dividend, cutting it can be costly in reputation or signaling terms. A manager facing a changing product market has an incentive to avoid this commitment.

Repurchases should also decline with fluidity, but the effect may be weaker because repurchases are more flexible and episodic than dividends. The paper indeed finds the strongest results for dividends, while still finding negative effects for repurchases.

3.2 Hypothesis 2: Product market fluidity and cash holdings

The second hypothesis is that competitive threats are positively related to cash holdings. Cash provides flexibility. Firms can use cash to respond to rival product moves, invest in new products, withstand margin pressure, or avoid external financing when external capital is expensive.

The prediction should be stronger for firms with less access to capital markets. If a firm can easily raise debt or equity at low cost, cash is less necessary as a buffer. If external finance is costly, cash becomes more valuable when product-market threats are high.

3.3 Life-cycle interpretation

The product life-cycle interpretation is that firms in stable product markets can more comfortably commit to payouts. Firms in fluid product markets are closer to a phase in which product designs, competitors, and customer demand are still evolving. Such firms retain flexibility rather than distribute cash.

This differs from the standard firm life-cycle interpretation. The paper's point is not only that old firms pay dividends. It is that firms facing stable product markets pay dividends, even after controlling for age and retained earnings.

3.4 Agency interpretation

A standard agency view says payouts discipline managers who might waste free cash flow. Product market threats can act as an alternative discipline. If competition is already intense and threatening, managers may have less scope to waste cash, and the firm may have a real need for liquidity. Under this logic, fluidity can reduce the need for payouts while increasing the value of cash.

4 Measurement of product market fluidity

4.1 Product word vectors

Let J_t be the number of unique product words used in all firms' product descriptions in year t. For each firm i, define an ordered Boolean vector

$$W_{i,t} \in \{0,1\}^{J_t},$$

where element j equals one if firm i's product description uses word j and zero otherwise. Normalize this vector to unit length:

$$N_{i,t} = \frac{W_{i,t}}{\|W_{i,t}\|}.$$

Thus, $N_{i,t}$ records firm i's product vocabulary after removing the mechanical effect of text length.

4.2 Aggregate rival change vector

To measure product market change, the paper constructs the aggregate vector of changes in word usage by rivals. In the paper's notation,

$$D_{t-1,t} \equiv \left| \sum_j (W_{j,t} - W_{j,t-1}) \right|.$$

The vector $D_{t-1,t}$ is large in word dimensions where firms in the economy are adding or dropping product words. It captures where product descriptions are moving.

In the empirical implementation, the authors focus on words in a local product market dictionary, so the measure emphasizes changes by nearby rivals rather than unrelated economy-wide text changes.

4.3 Product market fluidity

The firm's product market fluidity is the cosine similarity between its own product vocabulary and the normalized rival-change vector:

$$\text{Product Market Fluidity}_{i,t} = \left\langle N_{i,t}, \frac{D_{t-1,t}}{\|D_{t-1,t}\|} \right\rangle.$$

Because both vectors are nonnegative, the value lies between zero and one. It is high when the words that describe firm i's products are also the words being added or dropped by other firms.

4.4 Why fluidity is a threat measure

The key feature is that fluidity is based on rival movement. A firm can have a stable product description and still face high fluidity if rivals are changing around the firm's product vocabulary. This is the main distinction between product market fluidity and self-fluidity.

A high value means the firm occupies a part of product space where competitors are actively changing product descriptions. Such change can represent entry, innovation, substitution, or shifting business models by rivals.

4.5 Self-fluidity

Self-fluidity measures the firm's own product change from year $t-1$ to year t:

$$\text{Self-Fluidity}_{i,t} = 1 - \langle N_{i,t}, N_{i,t-1} \rangle .$$

Self-fluidity is high when the firm's own product description changes. Product market fluidity is high when rivals change in the firm's product space. The two variables capture different economic forces.

4.6 Static concentration

The paper also includes a text-based Herfindahl-Hirschman index, denoted TNIC HHI. This is a static measure of product-market concentration. It is not the same as fluidity. In the summary statistics, fluidity is negatively correlated with HHI, meaning more competitive industries tend to be more fluid, but the correlation is far from perfect.

5 Data and variables

5.1 Sample construction

The authors begin with Compustat-CRSP firm-years from 1997 to 2008, remove regulated utilities and financial firms, and require minimum asset and book-value screens. The final sample includes 42,999 firm-year observations with Compustat, CRSP, and 10-K text data.

5.2 Payout measures

A firm is a dividend payer if dividends per share are positive. A firm is a repurchaser if purchases of common and preferred stock net of reductions in preferred stock are positive. The paper also studies large repurchasers, two-year repurchasers, dividend initiators, dividend increasers, dividend decreasers, and dividend omitters.

5.3 Cash measure

The cash-holdings variable is cash plus equivalents divided by assets. In the cash regressions, the paper adds controls from the cash literature, including foreign tax burden and industry acquisition intensity.

5.4 Control variables

The main controls include total risk, market-to-book, asset growth, income/assets, size percentile, firm age, negative earnings dummy, R&D intensity, retained earnings/assets, cash-flow risk, and sales growth. The regressions also use year fixed effects and, in many specifications, text-based FIC-300 industry fixed effects.

6 Main empirical findings

6.1 Fluidity is economically meaningful

The examples in Table I are intuitive. Low-fluidity firms include stable product firms and retailers. High-fluidity firms include gaming and communication firms in 1997 and biotechnology/pharmaceutical firms in 2008. These examples fit the idea that fluidity captures changing competitive landscapes.

Table II shows that payout behavior varies sharply across fluidity quintiles. The fraction of dividend payers declines from 48.7 percent in the least-fluid quintile to 9.1 percent in the most-fluid quintile. Repurchase propensity also declines, and cash/assets rises from 10.2 percent to 37.5 percent. The transition matrices show that fluidity is persistent, but also tends to move toward stability over time, consistent with product life-cycle dynamics.

6.2 Fluidity captures entrepreneurial threats

The paper validates fluidity by comparing it to product descriptions of IPO firms and venture-capital-backed firms. Fluidity is strongly positively related to both IPOscore and VCscore. This supports the interpretation that fluidity captures emerging competitive threats.

Importantly, the coefficient on fluidity is much larger than the coefficient on risk in these regressions. This suggests that text-based fluidity contains threat information not captured by standard risk variables.

6.3 Dividend propensity

The dividend regressions show that firms in fluid product markets are less likely to pay dividends. The relation holds in logit and linear probability models, with and without extra controls and industry fixed effects.

In economic terms, moving local product market fluidity from its median to the 90th percentile reduces the predicted propensity to pay dividends from about 14.9 percent to about 8–9 percent, depending on the fixed effects specification.

6.4 Dividend levels and changes

Among dividend-paying firms, fluidity is associated with lower dividend yields and lower dividends/assets. Fluidity also predicts dynamic payout changes: fluid firms are less likely to initiate dividends and more likely to omit dividends. Firms that increase dividends tend to be in less-fluid product markets.

These results support the interpretation that firms require product-market stability before committing to dividend payments.

6.5 Repurchases

The repurchase results are weaker than the dividend results but point in the same direction. Firms in more fluid product markets are less likely to repurchase shares. This is true for the general repurchase dummy, large repurchases, and two-year repurchasers.

The weaker effect is natural because repurchases are less binding than dividends. Firms can repurchase opportunistically without establishing a long-term commitment.

6.6 Cash holdings

Fluidity is positively related to cash/assets. This relation is significant after including standard cash-holding controls and industry fixed effects. Firms in more fluid product markets hold more cash.

The effect is larger for firms that likely face higher external financing frictions: young firms, loss-making firms, and noninvestment-grade firms. This is consistent with a precautionary flexibility motive.

7 Regression specifications and interpretation

7.1 Dividend probability specification

The main dividend-payer specification can be summarized as

$$\Pr(\text{Dividend Payer}_{i,t} = 1) = \Lambda\left(\alpha_t + \beta\,\text{Fluidity}_{i,t} + \gamma' X_{i,t}\right),$$

where $\Lambda(\cdot)$ is the logit link, α_t are year effects, and $X_{i,t}$ includes risk, growth, profitability, size, age, R&D, retained earnings, and other controls. Some specifications add FIC-300 industry fixed effects.

The estimated β is negative. This means that, holding standard determinants fixed, product market threats reduce dividend propensity.

7.2 Payout level specification

For dividend-paying firms, the level regressions take the form

$$\text{Dividend Level}_{i,t} = \alpha_t + \beta\,\text{Fluidity}_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t},$$

where the dependent variable is either dividend yield or dividends/assets. The negative coefficient on fluidity indicates that even conditional on being a payer, firms facing greater product market threats pay less.

7.3 Dividend change specification

For initiations, omissions, increases, and decreases, the paper uses indicator outcomes. These dynamic outcomes help distinguish stable payout policy from changes in commitment. Fluidity is particularly important for initiations and omissions, which are direct transitions into or out of dividend-paying status.

7.4 Repurchase specification

The repurchase regressions mirror the dividend probability specification:

$$\Pr(\text{Repurchaser}_{i,t} = 1) = \Lambda\left(\alpha_t + \beta\,\text{Fluidity}_{i,t} + \gamma' X_{i,t}\right).$$

The coefficient on fluidity is generally negative. The result is robust across definitions of repurchase intensity, though the magnitudes are weaker than for dividends.

7.5 Cash specification

The cash regressions are OLS specifications of the form

$$\frac{\text{Cash}_{i,t}}{\text{Assets}_{i,t}} = \alpha_t + \beta\,\text{Fluidity}_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t}.$$

The estimated β is positive. The interpretation is precautionary: firms facing greater product market threats store more cash.

7.6 Why standardized coefficients matter

Many independent variables are standardized to unit standard deviation before estimation. Thus coefficients can be compared across regressors. A large coefficient on fluidity means that a one-standard-deviation increase in fluidity has a meaningful effect relative to conventional risk, size, growth, and profitability controls.

8 Robustness and identification concerns

8.1 Growth opportunities

A possible alternative is that fluidity simply proxies for growth options. The paper addresses this by controlling for market-to-book, asset growth, R&D/sales, patent measures, and firm age. The fluidity results remain.

8.2 Risk

A possible alternative is that fluidity is only a measure of cash-flow risk. The paper includes total risk and cash-flow risk, and shows that fluidity remains important. Fluidity and risk are related but not the same. The product text captures forward-looking threats that realized accounting risk may miss.

8.3 Firm maturity

A possible alternative is that fluidity just proxies for firm maturity. The paper includes age and retained earnings/assets. Fluidity continues to matter, supporting the claim that product market maturity is distinct from firm maturity.

8.4 Static competition

The paper controls for text-based HHI. The results show that both static concentration and dynamic fluidity matter. This is important because the economics of a stable concentrated industry differ from those of a dynamically fluid product market.

8.5 Endogeneity

One concern is that managers choose both product strategy and financial policy. The paper mitigates this concern because product market fluidity is based on changes by rival firms around the focal firm's product vocabulary. Thus it is less directly controlled by the focal firm's managers than self-fluidity.

9 Extracted figures and tables from the paper

9.1 Table I: Examples of Firms with High and Low Product Market Fluidity and Table II: Dividends, Repurchases, and Product Market Fluidity

Table I

Examples of Firms with High and Low Product Market Fluidity

In Panels A to D, we report firms that score in the lowest 25 or highest 25 based on their local product market fluidity in 1997 or 2008. In these panels, firms are sorted from most extreme to least extreme. In Panel A, we report firms that have the lowest local product fluidity (firms with stable products) based on their fiscal year 1997 Securities and Exchange Commission (SEC) filing. In Panel B, we report firms that have the highest local product fluidity (firms with stable products) based on their fiscal year 1997 SEC filing. In Panel C, we report firms that have the lowest local product fluidity (firms with stable products) based on their fiscal year 2008 SEC filing. In Panel D, we report firms that have the highest local product fluidity (firms with stable products) based on their fiscal year 2008 SEC filing.

Panel A: Firms in 1997 with Low Product Market Fluidity (Stable Products)
WEIS MARKETS, GENLYTE GROUP, ACME ELECTRIC, ALBERTO CULVER, SWANK, PHOENIX FOOTWEAR GROUP, BURKE MILLS, BALDOR ELECTRIC, FRANKLIN ELECTRIC, BUTLER MANUFACTURING, ROYAL APPLIANCE MFG, WYANT, WOLOHAN LUMBER, EASTMAN KODAK, MACYS, GOLD STANDARD, O SULLIVAN, LUFKIN INDUSTRIES, INTERNATIONAL ALUMINUM, MAY DEPARTMENT STORES, HARLAND JOHN H, ILLINOIS TOOL WORKS, NATIONAL SERVICE INDUSTRIES, GENESIS WORLDWIDE, SUPERIOR UNIFORM GROUP
Panel B: Firms in 1997 with High Product Market Fluidity (Fluid Products)
BOYD GAMING, COX COMMUNICATIONS, SHOWBOAT, COMCAST, TRUMP HOTELS & CASINO RESRTS, NEXTEL COMMUNICATIONS, P D L BIOPHARMA, MILLENNIUM PHARMACEUTICALS, ONYX PHARMACEUTICALS, HORIZON C M S HEALTHCARE, HARRAHS ENTERTAINMENT, LIGAND PHARMACEUTICALS, AMERISTAR CASINOS, TRIANGLE PHARMACEUTICALS, I D T, IMMUNE RESPONSE, SUN HEALTHCARE GROUP, HILTON HOTELS, MAGELLAN HEALTH SERVICES, NEXSTAR PHARMACEUTICALS, PLAYERS INTERNATIONAL, NEUREX, AZTAR, CORVAS INTERNATIONAL, MCLEODUSA
Panel C: Firms in 2008 with Low Product Market Fluidity (Stable Products)
SHERWIN WILLIAMS, ALBERTO CULVER, AMPCO PITTSBURGH, SUPERIOR UNIFORM GROUP, CINTAS, U S DATAWORKS, COLGATE PALMOLIVE, LIBERTY GLOBAL, COMPUTER SCIENCES, LAWSON PRODUCTS, VALHI, LIMITED BRANDS, PEPSIAMERICAS, ANIXTER INTERNATIONAL, E D A C TECHNOLOGIES, LANCE, FRIEDMAN INDUSTRIES, DECORATOR INDUSTRIES, MCCORMICK, FLEXSTEEL INDUSTRIES, STEPAN, PACCAR, CASS INFORMATION SYSTEMS, BLYTH, MOD PAC
Panel D: Firms in 2008 with High Product Market Fluidity (Fluid Products)
VICAL, ALTUS PHARMACEUTICALS, ALNYLAM PHARMACEUTICALS, ENZO BIOCHEM, ZYMOGENETICS, CYTOKINETICS, THRESHOLD PHARMACEUTICALS, ICAGEN, ANADYS PHARMACEUTICALS, INHIBITEX, ABRAXIS BIOSCIENCE, EMERGENT BIOSOLUTIONS, RIGEL PHARMACEUTICALS, G T X, OREXIGEN THERAPEUTICS, O S I PHARMACEUTICALS, AMGEN, UNITED THERAPEUTICS, ISIS PHARMACEUTI-CALS, IVIVI TECHNOLOGIES, BIOCRYST PHARMACEUTICALS, VERTEX PHARMACEUTICALS, ENZON PHARMACEUTICALS, NEUROGESX, I D M PHARMA

Table I: Examples of firms with high and low product market fluidity.

Table II

Dividends, Repurchases, and Product Market Fluidity

This table reports how dividends vary with local product market fluidity. Panel A displays summary payout statistics for the firms in each quintile based on local product market fluidity. Panels B to D present transition matrices examining ex post local product market fluidity for firms with varying levels of initial local product market fluidity for firms with at least 3 years as of a given year *t*. In Panel B, we examine the distribution of one year transition probabilities. In Panels C and D, we examine 3- and 6-year transition probabilities, respectively. For the 3-year test, we divide our 12-year sample into four nonoverlapping 3-year intervals. We then examine how product market fluidity changes for firms from one interval to the next. For the 6-year test, we divide our 12-year sample into two 6-year nonoverlapping intervals and repeat the same test. In each case, we assign firms to quintiles in each ex ante interval based on the level of ex ante product market fluidity. Holding break points fixed for each interval, we group observations into quintiles based on their ex post product market fluidity. The result is a 5×5 grid, in which we can compute the empirical distribution of transitions.

Sample	Most Stable	Quintile 2	Quintile 3	Quintile 4	Most Fluid	Obs.
Panel A: Payout Statistics						
% dividend payer	0.487	0.317	0.210	0.145	0.091	42,999
Dividend yield	0.012	0.006	0.004	0.003	0.002	42,999
Dividend/assets	0.011	0.006	0.004	0.003	0.002	42,999
% repurchasers	0.505	0.456	0.415	0.392	0.310	42,999
Cash+equiv/assets	0.102	0.139	0.191	0.248	0.375	42,999
Panel B: 1-Year Transition Probabilities						
Most stable product markets	0.778	0.168	0.040	0.010	0.005	4,691
Quintile 2	0.236	0.506	0.191	0.055	0.013	4,698
Quintile 3	0.034	0.268	0.460	0.188	0.050	4,697
Quintile 4	0.010	0.050	0.272	0.473	0.194	4,698
Most fluid product markets	0.004	0.009	0.039	0.237	0.711	4,693
Panel C: 3-Year Transition Probabilities						
Most stable product markets	0.779	0.177	0.032	0.010	0.003	1,597
Quintile 2	0.197	0.560	0.205	0.033	0.005	1,598
Quintile 3	0.031	0.240	0.506	0.202	0.022	1,599
Quintile 4	0.010	0.043	0.258	0.514	0.175	1,598
Most fluid product markets	0.001	0.004	0.051	0.223	0.721	1,598
Panel D: 6-Year Transition Probabilities						
Most stable product markets	0.745	0.197	0.051	0.005	0.002	589
Quintile 2	0.222	0.503	0.200	0.063	0.012	590
Quintile 3	0.048	0.284	0.448	0.178	0.042	589
Quintile 4	0.005	0.064	0.319	0.429	0.183	590
Most fluid product markets	0.002	0.015	0.075	0.284	0.625	589

Table II: Dividends, repurchases, and product market fluidity.

Read: Table I gives concrete high- and low-fluidity examples in 1997 and 2008. Table II shows payout and cash patterns across fluidity quintiles and reports fluidity transition probabilities.

Economic message: Stable-product firms are far more likely to pay dividends, while fluid-product firms hold much more cash. Fluidity is persistent but often declines over time, consistent with product markets maturing.

9.2 Table III: Fluidity and IPO/Venture Capital Product Descriptions and Table IV: Payout Summary Statistics

Table III

Fluidity and IPO/Venture Capital Product Descriptions

This table examines the effect of fluidity on the relation between a firm's product and the products of subsequent venture capital–financed firms and IPO firms. The dependent variable is IPOscore in columns 1 and 2 and VCscore in columns 3 and 4. IPOscore (VCscore) is equal to the average textual similarity between the given firm's 10-K business description in the given year, and the vocabulary of business descriptions extracted from SDC Platinum for firms going public (funded by venture capitalists) in the given year. The independent variables are all lagged and include lagged local product fluidity, total firm risk, the text-based industry HHI, the NYSE size percentile, and log firm age. Please see Table IV for a description of these variables. All independent variables are standardized prior to fitting regressions to permit more intuitive comparisons across variables. Observations are required to be in CRSP, Compustat, and our 10-K database. All specifications are estimated via OLS with year fixed effects. Specifications (2) and (4) include FIC-300 industry fixed effects. Robust standard errors are adjusted for heteroskedasticity and are clustered by firm. t-statistics are in parentheses.

	IPOscore (1)	IPOscore (2)	VCscore (3)	VCscore (4)
Total risk	−0.001	0.002	0.048	0.013
	(−1.249)	(0.224)	(4.506)	(1.520)
Local product fluidity	0.200	0.228	0.382	0.432
	(28.715)	(27.583)	(32.998)	(37.516)
R&D/sales	0.007	0.003	0.097	0.037
	(0.773)	(0.341)	(6.949)	(2.995)
Local fluidity X R&D/sales	−0.009	−0.012	−0.029	−0.022
	(−2.227)	(−3.132)	(−4.634)	(−3.991)
NYSE size percentile	0.066	0.069	0.075	0.083
	(8.769)	(10.253)	(6.159)	(8.564)
Log firm age	0.023	0.018	0.020	0.011
	(3.229)	(2.871)	(1.726)	(1.146)
HHI (TNIC)	−0.033	−0.001	−0.033	−0.033
	(−5.121)	(−1.469)	(−3.394)	(−3.791)
Constant	0.380	0.736	0.078	0.076
	(10.380)	(14.251)	(2.000)	(1.173)
Ind. Fixed Effects	No	Yes	No	Yes
R^2	0.204	0.262	0.195	0.405
N	35,489	35,489	35,489	35,489

Table III: Fluidity and IPO/venture capital product descriptions.

Table IV

Payout Summary Statistics

Summary statistics are reported for our sample of 42,999 observations based on annual firm observations from 1997 to 2008. Observations are required to be in CRSP, Compustat, and our 10-K database. We apply the same screens as Hoberg and Prabhala (2009), and the payout status variables are also analogously defined. To compute local product market fluidity, we first define the vector of aggregate absolute change in usage of each word in the product market universe from year $t-1$ to year t ($D_{t-1,t}$). Based on Hoberg and Phillips (2010b), we base this calculation only on the words in the local product market dictionary. A firm's local product market fluidity is then the cosine similarity between the given firm's normalized word usage vector in year t and $D_{t-1,t}$ (see the "Data" section for more details). This measures the extent to which product market words used by firm i are being adopted and dropped by other firms at a high rate. Self-fluidity is simply one minus the cosine similarity between firm i's year t product description and its year $t-1$ product description (higher values indicate that the firm is changing its own product composition). The HHI is the sales-weighted HHI of firms in a firm's industry. The variables in Panel C include other variables known to explain payer status as discussed in Appendix A.

Variable	Mean	*SD*	Minimum	Median	Maximum
Panel A: Data on Payout Status and Cash Holdings					
Dividend payer	0.245	0.430	0.000	0.000	1.000
Equity repurchaser	0.412	0.492	0.000	0.000	1.000
Both payer & repurchaser	0.157	0.364	0.000	0.000	1.000
Dividend yield	0.005	0.014	0.000	0.000	0.237
Dividend/assets	0.005	0.014	0.000	0.000	0.168
Dividend initiator	0.014	0.119	0.000	0.000	1.000
Dividend increaser	0.089	0.285	0.000	0.000	1.000
Dividend decreaser	0.007	0.082	0.000	0.000	1.000
Cash+equiv/assets	0.209	0.236	0.000	0.108	0.961
Panel B: Data from 10-K Text Analysis					
Local product market fluidity	6.932	3.362	1.374	6.379	20.628
Self product fluidity	21.043	14.920	1.457	17.154	78.464
HHI	0.221	0.231	0.019	0.127	1.000
Panel C: Data from the Existing Literature					
Total risk	0.043	0.025	0.010	0.037	0.168
Market-to-book	2.094	1.928	0.358	1.479	25.424
Asset growth	0.035	0.329	−3.196	0.055	0.875
Income/assets	−0.028	0.298	−3.001	0.061	0.325
NYSE size percentile	0.260	0.284	0.000	0.138	1.000
Log firm age	2.946	0.988	0.621	2.944	4.974
Negative earnings dummy	0.341	0.474	0.000	0.000	1.000
R&D/sales	0.329	1.676	0.000	0.002	23.816
Retained earnings/assets	−0.451	1.776	−15.439	0.080	0.914
Credit line dummy	0.612	0.487	0.000	1.000	1.000
Cash flow risk	0.337	0.053	0.211	0.329	0.615
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Table IV: Payout summary statistics.

Read: Table III validates fluidity by showing that it is strongly related to product descriptions of IPO and VC-backed firms. Table IV reports the main variable distributions.

Economic message: Fluidity captures entrepreneurial threats and not merely realized risk. The summary statistics also show that the sample contains substantial variation in payouts, cash, fluidity, risk, and firm maturity.

9.3 Table V: Pearson's Correlation Coefficients for Key 2008 Observations and Table VI: Dividend Policy and Product Fluidity

Table V

Pearson's Correlation Coefficients

We report Pearson's correlation coefficients for our sample of 42,999 observations based on annual firm observations from 1997 to 2008. Observations are required to be in CRSP, Compustat, and our 10-K database. See Table IV for a description of the variables.

Row	Variable	Local Product Market Fluidity	Self-Product Fluidity	HHI	Total Risk	Market-to -Book	Asset Growth	Income/ Assets	NYSE Percentile	Log Firm Age	R&D/ Sales
(1)	Self-fluidity	0.200									
(2)	HHI	−0.302	−0.021								
(3)	Total risk	0.177	0.191	0.045							
(4)	Market-to-book	0.245	0.104	−0.078	0.040						
(5)	Asset growth	−0.029	−0.018	−0.023	−0.262	0.120					
(6)	Income/assets	−0.312	−0.204	0.029	−0.470	−0.154	0.559				
(7)	NYSE size percentile	0.025	0.028	−0.191	−0.410	0.202	0.207	0.288			
(8)	Log firm age	−0.347	−0.136	0.031	−0.327	−0.130	0.025	0.267	0.279		
(9)	R&D/Sales	0.294	0.053	−0.076	0.109	0.174	−0.099	−0.342	−0.081	−0.126	
(10)	Neg. earn. dummy	0.306	0.194	−0.013	0.485	0.068	−0.334	−0.620	−0.341	−0.274	0.237

Table V: Pearson's correlation coefficients for key variables.

Table VI

Dividend Policy and Product Fluidity

Panel A reports logit estimates for our sample of 42,999 observations based on annual firm observations from 1997 to 2008. An observation is one firm in one year. The dependent variable is one if the firm is a dividend payer and zero otherwise. Observations are required to be in CRSP, Compustat, and our 10-K database. See Table IV for a description of the variables. All independent variables are standardized prior to fitting regressions to permit more intuitive comparisons across variables. In Panel B, we consider the subsample of dividend-paying firms, and the dependent variable is the percentage dividend yield or dividends/assets. Because these are continuous variables, we use OLS regressions, and we only report results for the specification with all controls to conserve space. In both panels, when the Extra Controls+Industry column is "Both," we add FIC-300 industry fixed effects and three additional variables (text+applied patents, the credit line dummy, and cash flow risk) but do not report coefficients to conserve space. All regressions are estimated with year fixed effects and standard errors are clustered by firm. t-statistics are in parentheses.

Row	Local Product Fluidity	Self-Product Fluidity	HHI	Total Risk	Log Firm Age	Market-to-Book	Asset Growth	Income/ Assets	NYSE Size Percentile	R&D/ Sales	Neg. Earn. Dummy	Ret Earnings/ Assets	3-Year Sales Growth	Extra Controls+ Industry	Num. Obs/ R^2
	Panel A: Dividend Payer vs. Nonpayer														
	Dependent variable: Dividend payer dummy (logistic model)														
(1)	−0.610	−0.051	0.068	−1.359	0.664	−0.374	−0.554	1.306	0.602					No	42,999
	(−12.98)	(−1.96)	(2.00)	(−18.83)	(15.54)	(−5.99)	(−16.27)	(12.85)	(14.12)						0.360
(2)	−0.416	−0.021	0.071	−1.117	0.566	−0.322	−0.377	0.826	0.652	−8.036	−0.032	2.174	−0.482	Controls	39,768
	(−8.23)	(−0.78)	(2.00)	(−15.07)	(12.62)	(−4.46)	(−10.60)	(5.60)	(14.46)	(−4.55)	(−0.36)	(7.61)	(−10.25)	Only	0.379
(3)	−0.363	0.023	0.021	−1.038	0.538	−0.161	−0.395	0.473	0.704	−1.125	−0.156	2.559	−0.503	Both	39,768
	(−6.31)	(0.82)	(0.55)	(−13.75)	(11.54)	(−2.37)	(−10.21)	(3.36)	(14.55)	(−0.95)	(−1.77)	(7.92)	(−10.20)		0.419
	Dependent variable: Dividend payer dummy (linear probability model)														
(4)	−0.076	−0.013	0.008	−0.079	0.101	−0.022	−0.035	−0.003	0.124					No	42,999
	(−17.78)	(−5.03)	(1.94)	(−21.79)	(20.76)	(−7.45)	(−15.19)	(−1.03)	(22.14)						0.347
(5)	−0.072	−0.011	0.010	−0.078	0.104	−0.022	−0.036	−0.012	0.125	0.016	−0.058	0.010	−0.018	Controls	39,768
	(−14.25)	(−4.03)	(2.30)	(−20.06)	(19.03)	(−6.44)	(−14.92)	(−3.37)	(21.09)	(8.47)	(−9.65)	(3.33)	(−8.25)	Only	0.354
(6)	−0.055	−0.002	−0.006	−0.064	0.091	−0.014	−0.036	−0.0133	0.123	0.009	−0.024	0.015	−0.021	Both	39,768
	(−10.44)	(−0.68)	(−1.41)	(−17.76)	(17.23)	(−4.31)	(−15.36)	(−4.05)	(21.35)	(4.26)	(−8.93)	(5.17)	(−10.04)		0.420
	Panel B: Dividend Levels														
	Dependent variable: Dividend yield														
(7)	−0.206	0.032	−0.007	0.127	0.081	−0.217	−0.710	−0.209	−0.243	−1.052	0.504	−1.174	−0.472	Both	10,343
OLS	(−3.73)	(1.25)	(−0.20)	(1.31)	(1.98)	(−3.43)	(−10.04)	(−0.94)	(−6.560)	(−0.80)	(3.30)	(−4.06)	(−6.42)		0.330
	Dependent variable: Dividend/assets														
(8)	−0.272	0.032	0.027	−0.40	0.01	1.209	−0.659	0.920	−0.141	1.588	0.159	−0.480	−0.465	Both	10,343
OLS	(−4.79)	(1.19)	(0.82)	(−5.61)	(0.37)	(14.16)	(−11.04)	(4.28)	(−3.54)	(1.13)	(1.274)	(−2.23)	(−6.04)		0.363

Table VI: Dividend policy and product fluidity.

Read: Table V shows that fluidity is correlated with HHI, age, R&D, and risk but not mechanically identical to them. Table VI shows that fluidity is negatively related to dividend payer status and dividend levels.

Economic message: Fluidity captures a distinct product-market state. Its dividend relation survives standard controls, supporting the financial-flexibility interpretation.

9.4 Table VII: Text Characteristics for Propensity Matched Firms and Table VIII: Economic Significance

Table VII

Text Characteristics for Propensity Matched Firms

We estimate a pooled logit explaining whether firms pay dividends using a panel from 1997 to 2008. Explanatory variables include nontext variables: total risk log firm age, firm size, M/B, asset growth, R&D/sales, negative earnings dummy, and year fixed effects. We group firms into four quartiles based on the predicted propensity to pay. The table reports the mean (first row) and median (second row) of three text-based product characteristics for payers and nonpayers for each quartile of the predicted propensity. We also test for the significance of the *differences between* payers and nonpayers within each quartile. Superscripts ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively, for the significance of these differences using t-tests or signed rank tests.

	Quartile 1		Quartile 2		Quartile 3		Quartile 4	
Variable	Payer	Nonpayer	Payer	Nonpayer	Payer	Nonpayer	Payer	Nonpayer
Local fluidity	6.695***	9.097	5.696***	7.157	5.493***	6.478	4.967***	6.101
	6.234***	8.776	5.250***	6.818	4.996***	5.998	4.393***	5.515
Self-fluidity	24.501	26.652	18.035***	21.616	17.212***	18.825	17.776	17.544
	18.622*	22.565	14.107***	17.884	13.542***	15.480	14.219	14.469
HHI	0.300***	0.205	0.286***	0.224	0.264***	0.211	0.211	0.211
	0.176***	0.102	0.189***	0.132	0.172***	0.133	0.132*	0.127
# observations	83	10,673	710	10,045	2,479	8,276	7,276	3,479

Table VII: Text characteristics for propensity matched firms.

Table VIII

Economic Significance

Panels A and B display economic effects for the propensity to pay dividends based on logit coefficient estimates from Table VI, specifications (2) and (3). The dependent variable is one if the firm is a dividend payer and zero otherwise. Each column gives the propensity to pay at the 10th, 25th, 50th, 75th, and 90th percentile of a variable holding the other variables at their median values. Observations are required to be in CRSP, Compustat, and our 10-K database. See Table IV for a description of the variables.

Percentile	Local Product Fluidity	Self-Product Fluidity	HHI	Total Risk	Log Firm Age	R&D/ Sales	Ret. Earnings	Cash Flow Risk	Sales Growth
	Panel A: Propensity to Pay Dividends: Full Model, No FIC Fixed Effects								
10	0.209	0.151	0.145	0.283	0.075	0.150	0.015	0.146	0.199
25	0.182	0.150	0.146	0.227	0.108	0.150	0.092	0.147	0.170
50	0.149	0.149	0.149	0.149	0.149	0.149	0.149	0.149	0.149
75	0.113	0.147	0.155	0.074	0.202	0.100	0.189	0.150	0.122
90	0.083	0.144	0.165	0.029	0.280	0.039	0.232	0.153	0.090
	Panel B: Propensity to Pay Dividends + FIC Fixed Effects								
10	0.210	0.151	0.146	0.286	0.083	0.149	0.020	0.146	0.200
25	0.184	0.150	0.147	0.232	0.114	0.149	0.100	0.147	0.171
50	0.149	0.149	0.149	0.149	0.149	0.149	0.149	0.149	0.149
75	0.117	0.147	0.155	0.080	0.207	0.102	0.192	0.151	0.123
90	0.087	0.145	0.166	0.033	0.285	0.045	0.235	0.154	0.091

Table VIII: Economic significance.

Read: Table VII compares payers and nonpayers within predicted propensity quartiles formed without text variables. Table VIII translates logit estimates into predicted dividend probabilities at different percentiles of each variable.

Economic message: Even among firms with similar nontext predicted payout propensity, payers have lower fluidity. The economic effect of fluidity on dividend propensity is large.

9.5 Table IX: Dividend Initiations, Omissions, Increases, and Decreases and Table X: Repurchase Policy and Product Fluidity

Table IX

Dividend Initiations, Omissions, Increases, and Decreases

The table presents results of logistic regressions for our sample based on annual firm observations from 1997 to 2008. An observation is one firm-year. In Panel A, the dependent variable is one if the firm is a dividend initiator and zero otherwise. In Panel A, the sample is limited to firms paying no dividends in year $t-1$ (32,658 observations). In Panel B, the dependent variable is one if the firm is a dividend omitter and zero otherwise. In Panels C and D, the dependent variable is one if the firm is a dividend increaser and decreaser, respectively. The sample in Panels B to D is limited to firms paying positive dividends in year $t-1$ (10,341 observations). Observations are required to be in CRSP, Compustat, and our 10-K database. See Table IV for a description of the variables. When the Extra Controls+Industry Effects column is "Yes," then industry fixed effects (based on SIC-3 or FIC-300 as noted) and three additional variables (text+applied patents, Bates, Kahle, and Stulz (2009) cash flow risk, and the credit line dummy) are included in the regression but not displayed to conserve space. All independent variables are standardized prior to fitting regressions to permit more intuitive comparisons across variables. Panel regressions are estimated with year fixed effects and standard errors are clustered by firm. t-statistics are in parentheses.

Row	Local Product Fluidity	Self-Product Fluidity	HHI	Total Risk	Log Firm Age	Market-to-Book	Asset Growth	Income/Assets	NYSE Size Percentile	R&D/Sales	Neg. Earn. Dummy	Retained Earnings /Assets	3-Year Sales Growth	Extra Controls+ Industry Effects	Obs./Pseudo-R^2
							Panel A: Dividend Initiations								
(1)	−0.397	−0.014	0.034	−0.666	0.029	−0.176	−0.777	1.660	0.181					Neither	32,658
	(−6.90)	(−0.28)	(0.79)	(−6.73)	(0.65)	(−2.16)	(−12.62)	(8.32)	(3.90)						0.028
(2)	−0.240	0.017	0.022	−0.559	0.032	−0.005	−0.717	1.098	0.258	−0.046	−0.269	0.161	−0.385	Both	29,620
	(−3.19)	(0.30)	(0.41)	(−5.33)	(0.70)	(−0.06)	(−9.13)	(3.91)	(4.97)	(−0.35)	(−2.67)	(1.26)	(−4.28)		0.039
							Panel B: Dividend Omissions								
(3)	0.314	0.164	0.068	0.543	−0.170	−0.265	−0.067	−0.195	−0.766					Neither	10,341
	(5.19)	(3.41)	(1.18)	(9.36)	(−3.03)	(−2.37)	(−1.26)	(−3.54)	(−7.80)						0.075
(4)	0.190	0.123	0.035	0.493	−0.171	−0.319	−0.027	0.052	−0.893	0.202	0.135	−0.486	−0.220	Both	10,148
	(2.12)	(2.18)	(0.51)	(7.53)	(−2.50)	(−2.28)	(−0.46)	(0.65)	(−7.79)	(1.63)	(2.27)	(−3.82)	(−3.55)		0.104
							Panel C: Dividend Increases								
(5)	−0.208	−0.165	−0.033	−0.594	0.086	0.223	0.138	0.284	0.383					Neither	10,341
	(−4.33)	(−5.33)	(−0.85)	(−9.55)	(2.06)	(5.16)	(4.85)	(5.69)	(8.31)						0.172
(6)	−0.151	−0.147	−0.069	−0.488	0.059	0.192	0.081	0.208	0.520	−0.280	−0.149	0.336	0.094	Both	10,148
	(−2.72)	(−4.68)	(−1.54)	(−7.68)	(1.32)	(3.78)	(2.58)	(3.15)	(9.56)	(−3.53)	(−3.45)	(5.30)	(2.58)		0.241
							Panel D: Dividend Decreases								
(7)	−0.100	0.193	−0.099	0.141	0.178	−0.167	−0.261	−0.107	−0.223					Neither	10,341
	(−1.29)	(3.73)	(−1.73)	(2.47)	(2.20)	(−0.99)	(−3.72)	(−1.54)	(−2.27)						0.022
(8)	−0.163	0.215	−0.161	0.055	0.173	−0.067	−0.151	−0.012	−0.152	−0.199	0.326	−0.114	−0.299	Both	10,148
	(−1.59)	(3.51)	(−2.00)	(0.81)	(2.19)	(−0.53)	(−2.07)	(−0.16)	(−1.49)	(−1.66)	(4.96)	(−1.47)	(−4.31)		0.047

Table IX: Dividend initiations, omissions, increases, and decreases.

Table X

Repurchase Policy and Product Fluidity

This table presents results of logistic regressions for our sample of 42,999 observations based on annual firm observations from 1997 to 2008. An observation is 1 firm-year. The dependent variable is one if the firm is a repurchaser and zero otherwise. In Panel B, the repurchaser dummy is further restricted to larger repurchases (at least 1% of assets), and in Panel C, the repurchaser dummy is further restricted to firms that repurchase both in the current year and in the previous year. Observations are required to be in CRSP, Compustat, and our 10-K database. See Table IV for a description of the variables. When the Extra Controls+Industry Effects column is "Yes," then industry fixed effects (based on SIC-3 or FIC-300 as noted) and three additional variables (text+applied patents, BKS cash flow risk, and the credit line dummy) are included in the regression but not displayed to conserve space. All independent variables are standardized prior to fitting regressions to permit more intuitive comparisons across variables. Panel regressions are estimated with year fixed effects and standard errors are clustered by firm. t-statistics are in parentheses.

Row	Local Product Fluidity	Self-Product Fluidity	TNIC HHI	Total Risk	Log Firm Age	Market-to-Book	Asset Growth	Income/Assets	NYSE Size Percentile	R&D/Sales	Neg. Earn. Dummy	Retained Earnings /Assets	3-Year Sales Growth	Extra Controls+ Industry Effects	Obs./Pseudo-R^2
							Panel A: Positive Repurchase Dummy								
(1)	−0.159	0.026	−0.026	−0.377	0.062	−0.083	−0.430	0.408	0.386					Neither	42,999
	(−7.79)	(1.84)	(−1.50)	(−15.66)	(3.27)	(−4.40)	(−21.77)	(14.50)	(17.51)						0.137
(2)	−0.115	0.028	−0.026	−0.371	0.022	−0.081	−0.420	0.234	0.384	−0.060	−0.172	0.152	−0.107	Controls	39,768
	(−5.06)	(1.91)	(−1.48)	(−14.11)	(1.05)	(−3.71)	(−19.58)	(7.21)	(16.36)	(−2.14)	(−9.29)	(5.36)	(−6.11)	Only	0.141
(3)	−0.098	−0.007	−0.014	−0.391	0.053	−0.108	−0.413	0.242	0.407	0.011	−0.162	0.153	−0.111	Both	39,768
	(−3.65)	(−0.44)	(−0.72)	(−14.40)	(2.46)	(−4.74)	(−18.96)	(7.43)	(17.08)	(0.41)	(−8.69)	(5.48)	(−6.17)		0.167
							Panel B: Repurchase More than 1% of Assets Dummy								
(4)	−0.131	0.018	−0.029	−0.429	−0.009	0.091	−0.569	0.531	0.394					Neither	42,999
	(−6.12)	(1.13)	(−1.52)	(−15.07)	(−0.47)	(5.02)	(−23.99)	(15.06)	(17.86)						0.124
(5)	−0.103	0.017	−0.041	−0.411	−0.061	0.086	−0.568	0.341	0.407	−0.056	−0.219	0.165	−0.146	Controls	39,768
	(−4.32)	(1.06)	(−2.09)	(−13.51)	(−2.90)	(4.07)	(−21.61)	(8.28)	(17.39)	(−1.58)	(−10.18)	(4.76)	(−6.87)	Only	0.134
(6)	−0.084	−0.027	−0.015	−0.458	−0.019	0.022	−0.580	0.380	0.443	0.019	−0.210	0.152	−0.140	Both	39,768
	(−3.00)	(−1.62)	(−0.75)	(−14.00)	(−0.87)	(0.99)	(−21.12)	(8.96)	(18.50)	(0.54)	(−9.62)	(4.58)	(−6.25)		0.162
							Panel C: 2-Year Repurchase Dummy								
(7)	−0.201	−0.005	−0.026	−0.539	0.146	−0.040	−0.447	0.452	0.403					Neither	42,999
	(−8.13)	(−0.31)	(−1.25)	(−16.08)	(6.72)	(−1.76)	(−20.05)	(12.04)	(16.18)						0.158
(8)	−0.145	−0.003	−0.031	−0.513	0.075	−0.012	−0.387	0.198	0.411	−0.081	−0.183	0.256	−0.266	Controls	39,768
	(−5.29)	(−0.16)	(−1.48)	(−14.47)	(3.17)	(−0.49)	(−16.16)	(4.61)	(15.69)	(−1.73)	(−8.09)	(5.19)	(−10.62)	Only	0.165
(9)	−0.114	−0.038	−0.025	−0.535	0.113	−0.051	−0.382	0.217	0.440	0.001	−0.176	0.248	−0.270	Both	39,768
	(−3.55)	(−2.15)	(−1.08)	(−14.45)	(4.53)	(−1.92)	(−15.69)	(5.10)	(16.39)	(0.01)	(−7.73)	(5.36)	(−10.53)		0.191

Table X: Repurchase policy and product fluidity.

Read: Table IX shows that fluidity predicts dividend transitions, especially initiations and omissions. Table X shows that fluidity is also negatively related to repurchases.

Economic message: Firms initiate or raise dividends when product markets are stable, and avoid or stop dividends when product threats rise. Repurchases are also lower under threat, though the relation is weaker than for dividends.

9.6 Table XI: Cash Holdings and Fluidity and Table XII: Cash Holdings and Fluidity: Age, Earnings, and Bond Ratings

Table XI

Cash Holdings and Fluidity

This table examines the effect of fluidity on cash holdings. OLS regressions are based on annual firm observations from 1997 to 2008. An observation is 1 firm-year. The dependent variable is equal to firm cash and cash equivalents divided by firm assets. Observations are required to be in CRSP, Compustat, and our 10-K database. In addition to the base variables defined in Table IV, we include an estimate of the foreign tax burden and industry acquisition intensity. Foreign tax burden equals the maximum of zero or foreign income times a firm's marginal effective tax rate computed as in Graham (1996) minus foreign taxes paid as in Foley et al. (2007). Industry acquisition intensity is the total number of acquisitions divided by the number of firms in a given industry. All independent variables are standardized prior to fitting regressions to permit more intuitive comparisons across variables. Column 1 includes year fixed effects while column 2 includes both year and FIC-300 industry fixed effects. Standard errors are clustered by firm and corrected for heteroskedasticity. *t*-statistics are in parentheses.

	(1)	(2) With Industry Controls
Local product fluidity	0.045	0.030
	(20.378)	(12.932)
TNIC HHI	−0.023	−0.011
	(−12.445)	(−5.589)
Log firm age	−0.011	−0.009
	(−5.650)	(−5.499)
Market-to-book	0.049	0.038
	(27.237)	(21.707)
ln(book assets)	−0.047	−0.042
	(−23.282)	(−20.729)
Earnings/assets	−0.006	−0.001
	(−2.985)	(−0.559)
Ind. acq. intensity	0.007	−0.001
	(4.715)	(−0.710)
Foreign tax burden	0.011	0.006
	(7.115)	(4.505)
Cash flow risk	0.033	0.013
	(14.531)	(5.831)
Capx/assets	−0.034	−0.024
	(−25.583)	(−17.377)
Neg. earn. dummy	0.023	0.006
	(6.183)	(1.666)
R&D/assets	0.056	0.043
	(20.151)	(13.785)
Constant	0.170	0.182
	(23.062)	(12.172)
R^2	0.441	0.525
[illegible]	[illegible]	[illegible]

Table XI: Cash holdings and fluidity.

Table XII

Cash Holdings and Fluidity: Age, Earnings, and Bond Ratings

This table examines the effect of fluidity on cash holdings by young versus old firms (columns 1 and 2), loss-making versus profitable firms (columns 3 and 4), and investment grade versus noninvestment grade firms (columns 5 and 6). OLS regressions are based on annual firm observations from 1997 to 2008. An observation is 1 firm-year. The dependent variable is equal to firm cash and cash equivalents divided by firm assets. Observations are required to be in CRSP, Compustat, and our 10-K database. In addition to the base variables defined in Table IV, we include an estimate of the foreign tax burden and industry acquisition intensity. Foreign tax burden equals the maximum of zero or foreign income times a firm's marginal effective tax rate computed as in Graham (1996) minus foreign taxes paid as in Foley et al. (2007). Industry acquisition intensity is the total number of acquisitions divided by the number of firms in a given industry. All independent variables are standardized prior to fitting regressions to permit more intuitive comparisons across variables. All specifications include year and FIC-300 industry fixed effects. Standard errors are clustered by firm and corrected for heteroskedasticity. *t*-statistics are in parentheses.

	(1) Young Firms	(2) Old Firms	(3) Neg. Earn.	(4) Pos. Earn.	(5) Noninvestment Grade	(6) Investment Grade
Local product fluidity	0.048	0.026	0.033	0.021	0.034	0.010
	(8.228)	(10.856)	(8.633)	(8.666)	(12.557)	(2.910)
TNIC HHI	−0.019	−0.010	−0.008	−0.009	−0.009	−0.009
	(−3.585)	(−4.971)	(−2.385)	(−4.727)	(−4.032)	(−2.916)
Log firm age	−0.011	−0.014	−0.023	−0.004	−0.010	−0.004
	(−1.299)	(−5.918)	(−7.013)	(−2.036)	(−5.034)	(−1.253)
Market-to-book	0.032	0.039	0.026	0.047	0.034	0.049
	(8.080)	(20.576)	(11.000)	(18.604)	(17.886)	(12.699)
ln(book assets)	−0.059	−0.040	−0.034	−0.043	−0.041	−0.041
	(−10.481)	(−18.450)	(−9.394)	(−20.844)	(−16.065)	(−11.209)
Earnings/assets	−0.001	0.001	−0.002	0.015	0.001	−0.001
	(−0.183)	(0.232)	(−0.742)	(1.859)	(0.225)	(−1.039)
Ind. acq. intensity	−0.0113	−0.0004	−0.0017	0.0012	−0.0012	0.0029
	(−2.075)	(−0.293)	(−0.845)	(0.747)	(−0.794)	(1.234)
Foreign tax burden	0.009	0.006	0.001	0.006	0.007	0.005
	(2.183)	(4.165)	(0.028)	(4.620)	(3.778)	(3.680)
Cash flow risk	0.014	0.013	0.020	0.008	0.017	0.003
	(2.308)	(5.428)	(4.909)	(3.541)	(6.065)	(0.921)
Capx/assets	−0.021	−0.025	−0.028	−0.022	−0.024	−0.023
	(−7.258)	(−16.373)	(−13.110)	(−13.837)	(−15.024)	(−9.156)
Neg. earn. dummy	0.024	0.004			0.002	0.004
	(2.642)	(1.092)			(0.447)	(0.387)
R&D/assets	0.021	0.049	0.032	0.394	0.039	0.292
	(4.147)	(13.544)	(11.181)	(2.934)	(12.965)	(2.345)
Constant	0.143	0.184	0.154	0.238	0.181	0.225
	(2.853)	(11.518)	(5.966)	(9.323)	(10.785)	(6.084)
R^2	0.548	0.528	0.537	0.459	0.532	0.493
N	4,477	37,160	14,112	27,525	30,438	11,199

Table XII: Cash holdings and fluidity by age, earnings, and bond ratings.

Read: Table XI shows that fluidity predicts higher cash/assets. Table XII shows that this effect is larger for young firms, negative-earnings firms, and noninvestment-grade firms.

Economic message: The cash evidence is the mirror image of the payout evidence. Product market threats induce firms to retain financial flexibility, especially when external financing access is limited.

10 Short summary

- The paper studies whether product market threats affect dividends, repurchases, and cash holdings.
- The key new measure is product market fluidity, constructed from annual 10-K product descriptions.
- Fluidity is high when rival firms are adding or dropping product words that overlap with the focal firm's products.
- Fluidity differs from self-fluidity: a firm can face high rival threat even if its own product description is stable.
- High-fluidity firms are less likely to pay dividends and pay lower dividend amounts.
- High-fluidity firms are less likely to repurchase shares, though the repurchase effects are weaker than the dividend effects.
- High-fluidity firms hold more cash.
- The cash effect is strongest for firms with less access to capital markets.
- Fluidity is strongly related to IPO and VC-backed product descriptions, supporting the threat interpretation.
- The results survive controls for risk, growth opportunities, firm maturity, static competition, and industry fixed effects.

11 Conclusion

Hoberg, Phillips, and Prabhala show that product market dynamics are a major determinant of corporate financial flexibility. Their text-based measure of product market fluidity captures the extent to which rivals are changing in a firm's product space. This measure predicts lower payouts and higher cash holdings.

The paper's strongest message is that product market threats are forward-looking and dynamic. Historical cash-flow volatility, firm age, and industry concentration do not fully capture them. A firm can be profitable and mature, but still face a fluid and threatening product market if rivals are changing rapidly around its products.

The findings help connect corporate finance and industrial organization. Firms do not set payout and cash policies only in response to balance-sheet variables. They also respond to the product market environment in which future cash flows and investment needs are created. When that environment is fluid, firms retain cash and avoid payout commitments.